APC 523 PSET 4

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April 2026

1 Elliptic BVP

Let Ω be the planar domain $[0,1]^2$. Consider the boundary value problem (BVP)

$$\left\{ \begin{array}{c} \nabla^2 u - u^4 = 0, x \in \Omega, \\ u(x) = 1, x \in \partial\Omega \end{array} \right\} \tag{1}$$

Ω is discretized into an $N \times N$ regular grid, with the field u taking values at

$$u_{i,j} = u(x_i, y_i) = u\left(\frac{i}{N}, \frac{j}{N}\right), 0 \leq i, j \leq N$$

We approximate the Laplace operator using the classical 5-point second-order finite-difference approximation.

$$\Delta_h u_{i,j} = \frac{u_{i-1,j} + u_{i,j-1} - 4u_{i,j} + u_{i+1,j} + u_{i,j+1}}{h^2} = \Delta u(x_i, y_i) + \mathcal{O}(h^2)$$

(a) *Reformulate the boundary value problem as a root-finding problem that can be solved using the Newton-Raphson method.*

First, we can rewrite (1) as

$$f(\mathbf{u}) = \left\{ \begin{array}{ll} \frac{u_{i-1,j}+u_{i,j-1}-4u_{i,j}+u_{i+1,j}+u_{i,j+1}}{h^2} - (u_{i,j})^4 = 0 & \text{for } i,j \in [1, N-1], \\ 1 - u_{i,j} = 0 & \text{for } i,j = 0, N \end{array} \right. \tag{2}$$

For the Newton-Raphson method, we will then solve

$$\mathbb{J}(\mathbf{u}^{n+1} - \mathbf{u}^n) = -f(\mathbf{u}^n)$$

$$\therefore \mathbf{u}^{n+1} = \mathbf{u}^n - \mathbb{J}^{-1} f(\mathbf{u}^n) \tag{3}$$

where $\mathbb{J}$ is defined as

$$\frac{1}{h^2} \begin{bmatrix} 1 & 0 & 0 & \dots & & & 0 \\ 0 & 1 & 0 & \dots & & & 0 \\ \vdots & \ddots & \ddots & \ddots & & & \vdots \\ \vdots & 0 & 1 & -4 - 4h^2 u_{i,i}^3 & 1 & 0 & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & & & \dots & 0 & 0 & 1 \end{bmatrix} \tag{4}$$

with $\mathbb{J}_{(iN+j),(iN+j)} = 1/h^2$ where $i = 0, N-1$ or $j = 0, N-1$. Also, $\mathbb{J}_{(iN+j),(iN+j-1)} = 1/h^2$, $\mathbb{J}_{(iN+j),(iN+j+1)} = 1/h^2$, $\mathbb{J}_{(iN+j),((i-1)N+j)} = 1/h^2$, and $\mathbb{J}_{(iN+j),((i+1)N+j)} = 1/h^2$.

(b) *Implement the algorithm for method (a) computing directly the inverse of the Jacobian matrix using a standard linear algebra library. Plot the solution for $N = 64$. Be careful with the convergence criterion of the scheme.*

See `p1b.ipynb` for code implementation.

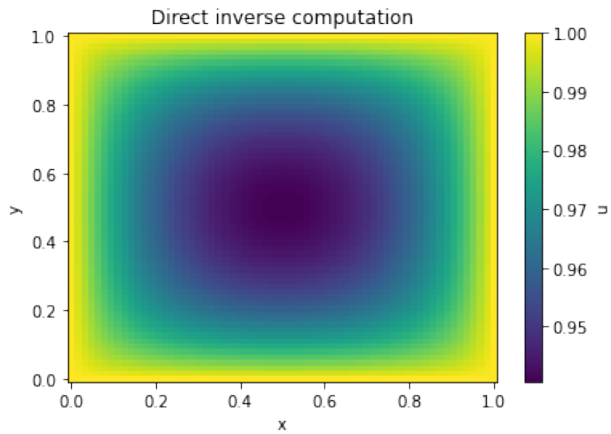


Figure 1: Newton-Raphson solve of (1) using `np.linalg.inv()` to evaluate Jacobian inverse ($N = 64$).

(c) *Avoid explicitly computing the inverse Jacobian by using instead a simple iterative method (e.g., Gauss-Seidel, Jacobi, etc.), with an initial guess for each Newton step given by the step before. Be careful with the convergence criterion of both the iterative matrix inversion and the Newton-Raphson method. Demonstrate that the method achieves second-order accuracy by comparing results for $N = 128$ and $N = 256$, using the solution at $N = 512$ as ground truth. You may use the infinity norm $||u||_\infty = \max |u_{i,j}|$, L^2 norm $||u||_2^2 = \frac{1}{(N+1)^2} \sum_{i,j} |u_{i,j}|^2$ or another sensible discrete norm to measure error.*

Code is run in `p1c.py`, saved to CSV files, and processed into figures in `p1c.ipynb`.

I used a Jacobi method with a strict update tolerance of $1e-9$ and a much looser Newton-Raphson tolerance of 0.1.

Indeed, we observe second-order accuracy, as reducing h from ≈ 0.008 to ≈ 0.004 yields a decrease in the infinity-norm from ≈ 0.04 to ≈ 0.0004 (that is, the infinity-norm is proportional to h^2).

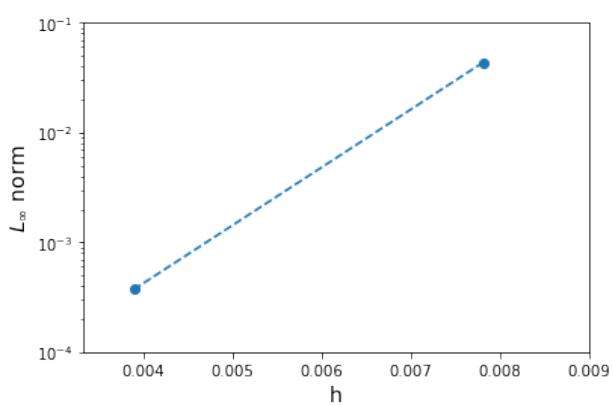
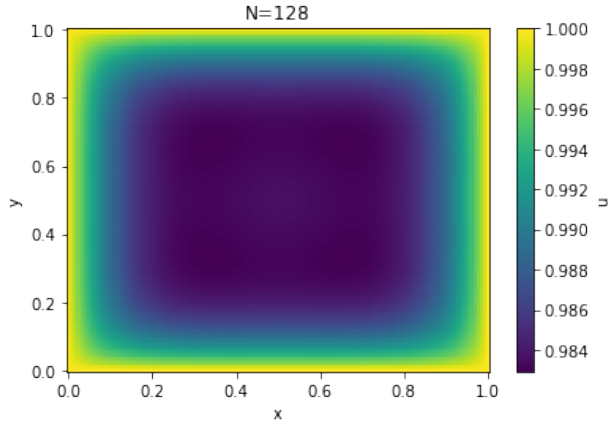
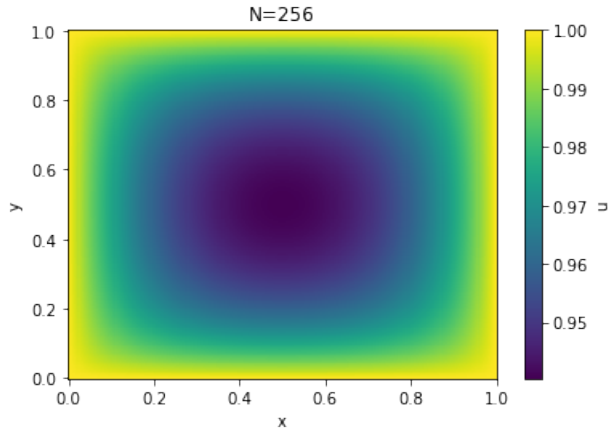


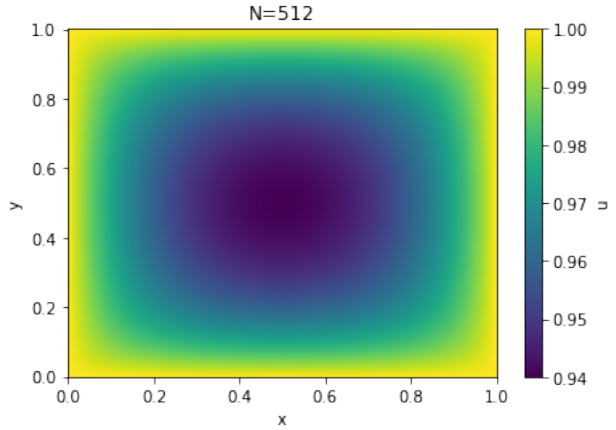
Figure 2: h^2 convergence for $N = 128, 256$ using Jacobi method.



(a)



(b)



(c)

Figure 3: Elliptic BVP solutions using iterative Jacobi method for (a) $N = 128$, (b) $N = 256$, and (c) $N = 512$.

2 Beam-warming Scheme

Derive the modified equation for the Beam-warming scheme presented in the lecture on "Hyperbolic PDEs":

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + a\frac{3u_i^n - 4u_{i-1}^n + u_{i-2}^n}{2h} = \frac{a^2\Delta t}{2}\frac{u_i^n - 2u_{i-1}^n + u_{i-2}^n}{h^2} \tag{5}$$

First, we can define each term in (5) with a Taylor-series expansion:

$$u_i^{n+1} = u_i^n + \Delta t\frac{\partial u}{\partial t} + \frac{\Delta t^2}{2!}\frac{\partial^2 u}{\partial t^2} + \frac{\Delta t^3}{3!}\frac{\partial^3 u}{\partial t^3} + \dots$$

$$u_{i-1}^n = u_i^n - h\frac{\partial u}{\partial x} + \frac{h^2}{2!}\frac{\partial^2 u}{\partial x^2} - \frac{h^3}{3!}\frac{\partial^3 u}{\partial x^3} + \dots$$

$$u_{i-2}^n = u_i^n - 2h\frac{\partial u}{\partial x} + \frac{4h^2}{2!}\frac{\partial^2 u}{\partial x^2} - \frac{8h^3}{3!}\frac{\partial^3 u}{\partial x^3} + \dots$$

$$u_{i+1}^n = u_i^n + h\frac{\partial u}{\partial x} + \frac{h^2}{2!}\frac{\partial^2 u}{\partial x^2} + \frac{h^3}{3!}\frac{\partial^3 u}{\partial x^3} + \dots$$

Using these expansions, we rewrite each part of (5) to third order:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{\partial u}{\partial t} + \frac{\Delta t}{2!}\frac{\partial^2 u}{\partial t^2} + \frac{\Delta t^2}{3!}\frac{\partial^3 u}{\partial t^3} \tag{6}$$

$$\begin{aligned} a\frac{3u_i^n - 4u_{i-1}^n + u_{i-2}^n}{2h} &= \frac{a}{2h}\left(4h\frac{\partial u}{\partial x} - \frac{4h^2}{2!}\frac{\partial^2 u}{\partial x^2} + \frac{4h^3}{3!}\frac{\partial^3 u}{\partial x^3} - 2h\frac{\partial u}{\partial x} + \frac{4h^2}{2!}\frac{\partial^2 u}{\partial x^2} - \frac{8h^3}{3!}\frac{\partial^3 u}{\partial x^3}\right), \\ &= a\left(\frac{\partial u}{\partial x} - \frac{1}{3}h^2\frac{\partial^3 u}{\partial x^3}\right) \end{aligned} \tag{7}$$

$$\begin{aligned} \frac{u_i^n - 2u_{i-1}^n + u_{i-2}^n}{h^2} &= \frac{1}{h^2}\left(2h\frac{\partial u}{\partial x} - \frac{2h^2}{2!}\frac{\partial^2 u}{\partial x^2} + \frac{2h^3}{3!}\frac{\partial^3 u}{\partial x^3} - 2h\frac{\partial u}{\partial x} + \frac{4h^2}{2!}\frac{\partial^2 u}{\partial x^2} - \frac{8h^3}{3!}\frac{\partial^3 u}{\partial x^3}\right), \\ &= \frac{\partial^2 u}{\partial x^2} - h\frac{\partial^3 u}{\partial x^3} \end{aligned} \tag{8}$$

From the original hyperbolic PDE equation, we have

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0$$

Thus, we can rewrite higher order time derivatives:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t}\left(-a\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left(-a\frac{\partial u}{\partial t}\right) = a^2\frac{\partial^2 u}{\partial x^2}$$

$$\text{(by same reasoning)} \ \rightarrow \frac{\partial^3 u}{\partial t^3} = -a^3\frac{\partial^3 u}{\partial x^3}$$

Combining the rewritten time-derivative terms with the expressions from (6), (7), and (8) gives

$$\frac{\partial u}{\partial t} + \frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \frac{\Delta t^2}{6}\frac{\partial^3 u}{\partial t^3} + a\left(\frac{\partial u}{\partial x} - \frac{h^2}{3}\frac{\partial^3 u}{\partial x^3}\right) = \frac{a^2\Delta t}{2}\left(\frac{\partial^2 u}{\partial x^2} - h\frac{\partial^3 u}{\partial x^3}\right)$$

$$\frac{\partial u}{\partial t} = a^3\frac{\Delta t^2}{6}\frac{\partial^3 u}{\partial x^3} - a\left(\frac{\partial u}{\partial x} - \frac{h^2}{3}\frac{\partial^3 u}{\partial x^3}\right) - \frac{a^2 h\Delta t}{2}\frac{\partial^3 u}{\partial x^3}$$

$$\begin{aligned} \therefore \frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} &= \frac{\partial^3 u}{\partial x^3}\left(\frac{a^3\Delta t^2}{6} + \frac{ah^2}{3} - \frac{a^2 h\Delta t}{2}\right), \\ &= \frac{\partial^3 u}{\partial x^3}\frac{ah^2}{6}\left(C^2 - 3C + 2\right) \text{ for } C = \frac{a\Delta t}{h} \end{aligned} \tag{9}$$

where (9) is the modified equation.

3 Advection

Consider a 2D advection equation for $u(t, x, y)$ given by

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} + b\frac{\partial u}{\partial y} = 0 \tag{10}$$

defined over a simulation domain of $\Omega = [0, 1]^2$ with double-periodic boundary conditions. You are given the initial condition

$$u(0, x, y) = \exp\left(-\frac{(x - 1/2)^2 + (y - 1/2)^2}{(3/20)^2}\right)$$

with advections speeds $a = 1$ and $b = 2$.

(a) *Show that the analytical solution is*

$$u(t, x, y) = u(0, x - at, y - bt). \tag{11}$$

Defining a change of variables:

$$\zeta = x - at,$$
$$\beta = y - bt$$

then

$$\frac{\partial \zeta}{\partial x} = 1, \frac{\partial \zeta}{\partial t} = -a, \frac{\partial \zeta}{\partial y} = 0$$

and

$$\frac{\partial \beta}{\partial y} = 1, \frac{\partial \beta}{\partial t} = -b, \frac{\partial \beta}{\partial x} = 0$$

Using the chain rule, we can thus write

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial \zeta}\frac{\partial \zeta}{\partial t} + \frac{\partial u}{\partial \beta}\frac{\partial \beta}{\partial \partial t}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \zeta}\frac{\partial \zeta}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \beta}\frac{\partial \beta}{\partial y}$$

Therefore, the 2D advection equation (10) can be rewritten as

$$\left(\frac{\partial u}{\partial \zeta}\frac{\partial \zeta}{\partial t} + \frac{\partial u}{\partial \beta}\frac{\partial \beta}{\partial \partial t}\right) + a\left(\frac{\partial u}{\partial \zeta}\frac{\partial \zeta}{\partial x}\right) + b\left(\frac{\partial u}{\partial \beta}\frac{\partial \beta}{\partial y}\right) = 0$$

$$-a\frac{\partial u}{\partial \zeta} - b\frac{\partial u}{\partial \beta} + a\frac{\partial u}{\partial \zeta} + b\frac{\partial u}{\partial \beta} = 0$$

which is necessarily true. Thus, the change of variables to (ζ, β) satisfies the PDE for all times t.

To satisfy the initial condition, the analytical solution must be

$$u(t, x, y) = u(0, x - at, y - bt)$$

(b) *We will now derive a 2D finite-volume numerical scheme based on this exact solution called the Corner Transport Upstream (CTU) method.*

(1) *For 1D advection, defining $\mu = a\Delta t/\Delta x$, we can reformulate the first-order upwind scheme using the exact solution for piecewise constant initial conditions, projected back to the initial mesh via conservative integration. Show geometrically (using a simple diagram) that projecting back the exact solution gives you the numerical solution*

$$u_i^{n+1} = (1 - \mu)u_i^n + \mu u_{i-1}^n$$

Note: I used [2] to guide my approach to this problem.
Figure 4 shows u_i^{n+1} projected back to the grid at $t = t^n$ along the characteristic line of advection, intersecting at $u_*^n(x_*)$.

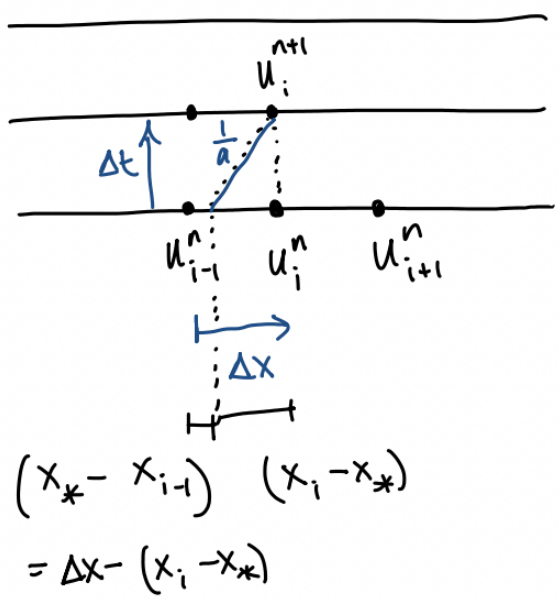


Figure 4: 1D advection sketch

We know that $u_i^{n+1} = u_*^n$, so if we can estimate u_*^n by interpolating from the known values at $t = t^n$, then we can also estimate u_i^{n+1}.
Geometrically,

$$x_* = x_i - a\Delta t$$

$$x_i - x_* = a\Delta t$$

We determine u_*^n by performing a weighted average between u_{i-1}^n and u_i^n.

$$\begin{aligned} u_*^n &= \frac{x_i - x_*}{\Delta x} u_{i-1}^n + \frac{\Delta x - (x_i - x_*)}{\Delta x} u_i^n, \\ &= \frac{a\Delta t}{\Delta x} u_{i-1}^n + \left(1 - \frac{a\Delta t}{\Delta x}\right) u_i^n, \\ &= \mu u_{i-1}^n + (1-\mu) u_i^n \end{aligned}$$

(2) *Using the same geometrical strategy in 2D, derive the projected values of the exact solution back to the initial grid as a function of $\mu \equiv a\Delta t/\Delta x$ and $\nu \equiv b\Delta t/\Delta y$. This should give you the fully discrete form of the CTU scheme.*

Note: I used [1] to help guide my approach to this problem.

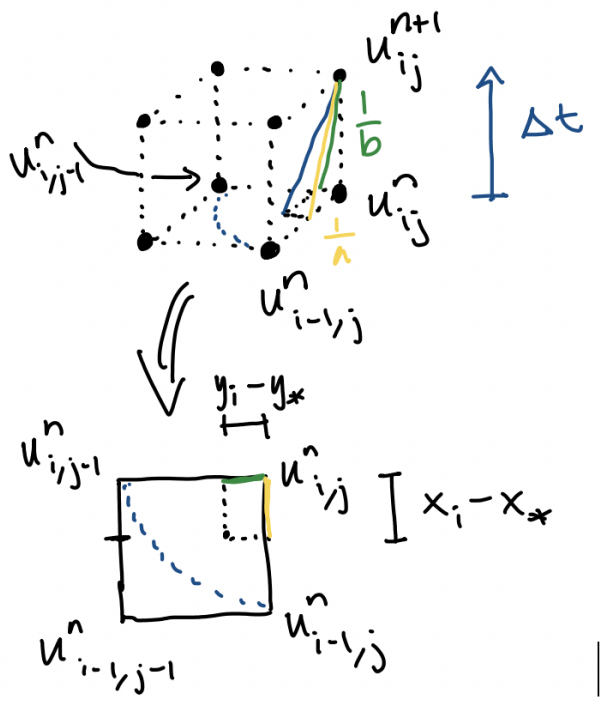


Figure 5: 2D advection sketch

In this case, the projected characteristic line from $u_{i,j}^{n+1}$ intersects the plane from $t = t^n$ at (x_*, y_*) such that

$$\begin{aligned} y_i - y_* &= b\Delta t, \\ x_i - x_* &= a\Delta t \end{aligned}$$

It is easier in this case to conceptualize the advection from $u_{*,*}^n$ to $u_{i,j}^{n+1}$ in intermediate steps: $u_{*,*}^n \to u_{*,j} \to u_{i,j}^{n+1}$.

$$\begin{aligned} u_{*,j} &= \frac{a\Delta t}{\Delta x} u_{i-1,j}^n + \left(1 - \frac{a\Delta t}{\Delta x}\right) u_{i,j}^n \\ &= \mu u_{i-1,j}^n + (1-\mu) u_{i,j}^n \end{aligned}$$

$$
\begin{aligned}
u_{i,j}^{n+1} &= \frac{b\Delta t}{\Delta y} u_{i,j-1}^n + \left(1 - \frac{b\Delta t}{\Delta y}\right) u_{*,j} \\
&= \nu u_{*,j-1} + (1-\nu)\left[\mu u_{i-1,j}^n + (1-\mu) u_{i,j}^n\right]
\end{aligned}
$$

Using

$$
u_{*,j-1} = \frac{a\Delta t}{\Delta x} u_{i-1,j-1}^n + \left(1 - \frac{a\Delta t}{\Delta x}\right) u_{i,j-1}^n
$$

then we get the CTU scheme:

$$
u_{i,j}^{n+1} = (1-\nu)(1-\mu)u_{i,j}^n + (1-\nu)\mu u_{i-1,j}^n + \nu(1-\mu)u_{i,j-1}^n + \nu\mu u_{i-1,j-1}^n
$$

(3) *Derive the modified equation of this scheme and analyze the leading-order error term.*

As before, we first expand each term of the fully discretized scheme equation by its Taylor series to third order:

$$
u_{i-1,j}^n = u_{i,j}^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2}\frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6}\frac{\partial^3 u}{\partial x^3}
$$

$$
u_{i,j-1}^n = u_{i,j}^n - \Delta y \frac{\partial u}{\partial y} + \frac{\Delta y^2}{2}\frac{\partial^2 u}{\partial y^2} - \frac{\Delta y^3}{6}\frac{\partial^3 u}{\partial y^3}
$$

$$
u_{i-1,j-1}^n = u_{i,j}^n - \Delta x \frac{\partial u}{\partial x} - \Delta y \frac{\partial u}{\partial y} + \frac{\Delta x^2}{2}\frac{\partial^2 u}{\partial x^2} + \Delta x \Delta y \frac{\partial^2 u}{\partial x \partial y} + \frac{\Delta y^2}{2}\frac{\partial^2 u}{\partial y^2} + \mathcal{O}(\Delta x^3 + \Delta y^3)
$$

$$
u_{i,j}^{n+1} = u_{i,j}^n + \Delta t \frac{\partial u}{\partial t} + \frac{\Delta t^2}{2}\frac{\partial^2 u}{\partial t^2} + \frac{\Delta t^3}{6}\frac{\partial^3 u}{\partial t^3}
$$

Using these expansions, we can rewrite the original equation as

$$
\begin{aligned}
u_{i,j}^n + \Delta t \frac{\partial u}{\partial t} + \frac{\Delta t^2}{2}\frac{\partial^2 u}{\partial t^2} + \frac{\Delta t^3}{6}\frac{\partial^3 u}{\partial t^3} = \\
&\nu\mu\left(u_{i,j}^n - \Delta x \frac{\partial u}{\partial x} - \Delta y \frac{\partial u}{\partial y} + \frac{\Delta x^2}{2}\frac{\partial^2 u}{\partial x^2} + \Delta x \Delta y \frac{\partial^2 u}{\partial x \partial y} + \frac{\Delta y^2}{2}\frac{\partial^2 u}{\partial y^2} + \mathcal{O}(\Delta x^3 + \Delta y^3)\right) \\
&+ \nu(1-\mu)\left(u_{i,j}^n - \Delta y \frac{\partial u}{\partial y} + \frac{\Delta y^2}{2}\frac{\partial^2 u}{\partial y^2} - \frac{\Delta y^3}{6}\frac{\partial^3 u}{\partial y^3}\right) \\
&+ (1-\nu)\mu\left(u_{i,j}^n - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2}\frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{6}\frac{\partial^3 u}{\partial x^3}\right) \\
&+ (1-\nu)(1-\mu)u_{i,j}^n
\end{aligned}
$$

Notice that the $u_{i,j}^n$ terms cancel on either side of the equation above.

We can rewrite the right-hand side as

$$
\begin{aligned}
=&\nu\mu\left(-\Delta x\frac{\partial u}{\partial x}-\Delta y\frac{\partial u}{\partial y}+\frac{\Delta x^2}{2}\frac{\partial^2 u}{\partial x^2}+\Delta x\Delta y\frac{\partial^2 u}{\partial x\partial y}+\frac{\Delta y^2}{2}\frac{\partial^2 u}{\partial y^2}+\mathcal{O}(\Delta x^3+\Delta y^3)\right)\\
&+\nu\left(-\Delta y\frac{\partial u}{\partial y}+\frac{\Delta y^2}{2}\frac{\partial^2 u}{\partial y^2}-\frac{\Delta y^3}{6}\frac{\partial^3 u}{\partial y^3}\right)\\
&-\mu\nu\left(-\Delta y\frac{\partial u}{\partial y}+\frac{\Delta y^2}{2}\frac{\partial^2 u}{\partial y^2}-\frac{\Delta y^3}{6}\frac{\partial^3 u}{\partial y^3}\right)\\
&+\mu\left(-\Delta x\frac{\partial u}{\partial x}+\frac{\Delta x^2}{2}\frac{\partial^2 u}{\partial x^2}-\frac{\Delta x^3}{6}\frac{\partial^3 u}{\partial x^3}\right)\\
&-\mu\nu\left(-\Delta x\frac{\partial u}{\partial x}+\frac{\Delta x^2}{2}\frac{\partial^2 u}{\partial x^2}-\frac{\Delta x^3}{6}\frac{\partial^3 u}{\partial x^3}\right)
\end{aligned}
$$

which simplifies to

$$
\begin{aligned}
=&\nu\mu\left(\Delta x\Delta y\frac{\partial^2 u}{\partial x\partial y}\right)\\
&+\nu\left(-\Delta y\frac{\partial u}{\partial y}+\frac{\Delta y^2}{2}\frac{\partial^2 u}{\partial y^2}-\frac{\Delta y^3}{6}\frac{\partial^3 u}{\partial y^3}\right)\\
&+\mu\left(-\Delta x\frac{\partial u}{\partial x}+\frac{\Delta x^2}{2}\frac{\partial^2 u}{\partial x^2}-\frac{\Delta x^3}{6}\frac{\partial^3 u}{\partial x^3}\right)
\end{aligned}
$$

Taking the time-derivative of the original equation (10), we see that

$$
\frac{\partial^2 u}{\partial t^2}+a\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial t}\right)+b\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial t}\right)=0
$$

$$
\begin{aligned}
\therefore\ \frac{\partial^2 u}{\partial t^2}&=-a\left(-a\frac{\partial^2 u}{\partial x^2}-b\frac{\partial^2 u}{\partial x\partial y}\right)-b\left(-a\frac{\partial^2 u}{\partial x\partial y}-b\frac{\partial^2 u}{\partial y^2}\right)\\
&=a^2\frac{\partial^2 u}{\partial x^2}+2ab\frac{\partial^2 u}{\partial x\partial y}+b^2\frac{\partial^2 u}{\partial y^2}
\end{aligned}
$$

Combining all expressions,

$$
\begin{aligned}
\Delta t\frac{\partial u}{\partial t}+\mu^2\frac{\Delta x^2}{2}\frac{\partial^2 u}{\partial x^2}+\nu\mu\Delta x\Delta y\frac{\partial^2 u}{\partial x\partial y}+\nu^2\frac{\Delta y^2}{2}\frac{\partial^2 u}{\partial y^2}&\\
=\nu\mu\Delta x\Delta y\frac{\partial^2 u}{\partial x\partial y}-\nu\Delta y\frac{\partial u}{\partial y}+\nu\frac{\Delta y^2}{2}\frac{\partial^2 u}{\partial y^2}-\mu\Delta x\frac{\partial u}{\partial x}+\mu\frac{\Delta x^2}{2}\frac{\partial^2 u}{\partial x^2}+\mathcal{O}(\Delta x^3+\Delta y^3)&
\end{aligned}
$$

Therefore, the modified equation is

$$
\frac{\partial u}{\partial t}+a\frac{\partial u}{\partial x}+b\frac{\partial u}{\partial y}=a(1-\mu)\frac{\Delta x}{2}\frac{\partial^2 u}{\partial x^2}+b(1-\nu)\frac{\Delta y}{2}\frac{\partial^2 u}{\partial y^2}+\mathcal{O}(\Delta x^3+\Delta y^3)
$$

with a $\Delta x^3, \Delta y^3$ leading-order error term.

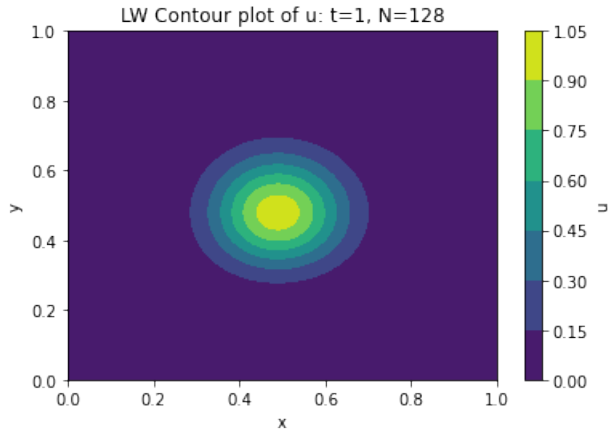
(c) *Evolve the system for* $t \in [0, 10]$ *discretized into* $N \times N$ *with* $N = \{128, 256, 512\}$ *using:*

(1) *the CTU method derived above*

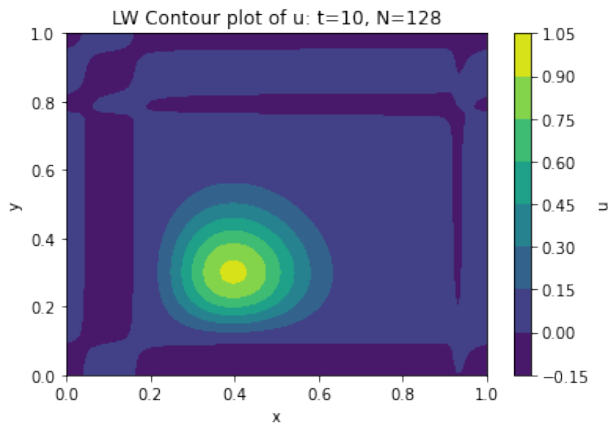
(2) *the Lax-Wendroff method*

For each case, plot the L_2 *norm of the error as a function of time and the contours of* $u(1, x, y)$ *and* $u(10, x, y)$ *for all resolutions.*

See figures on following pages. I sacrificed accuracy for runtime, as the job kept terminating prematurely on the Adroit cluster (running serially). The results suggest that these methods are likely poorly suited to this 2D advection problem, as the error trend does not improve for higher resolution, so the approach limits the accuracy of the solution.



(a)



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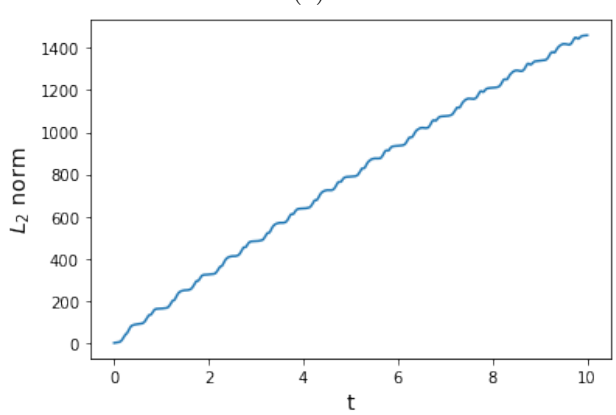
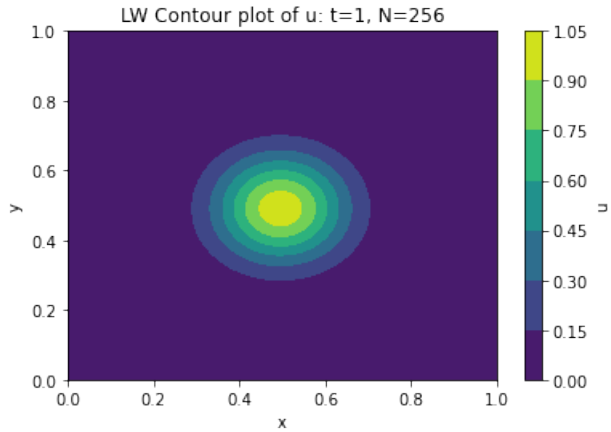
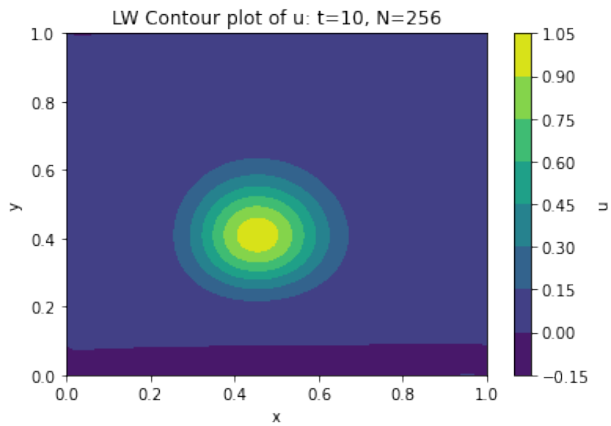


(c)

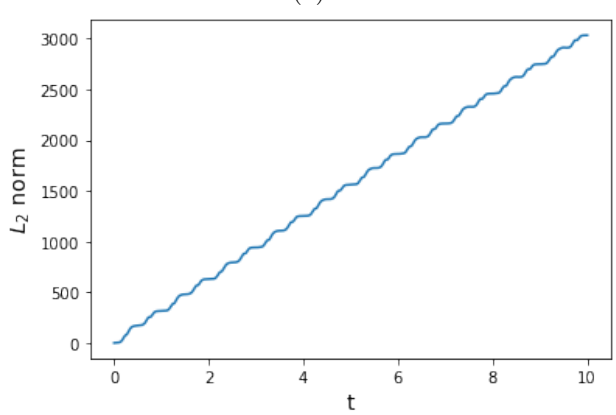
Figure 6: Lax-Wendroff method for $N = 128$



(a)

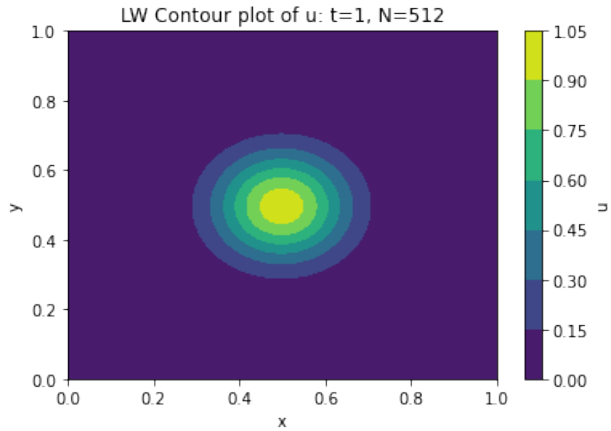


(b)

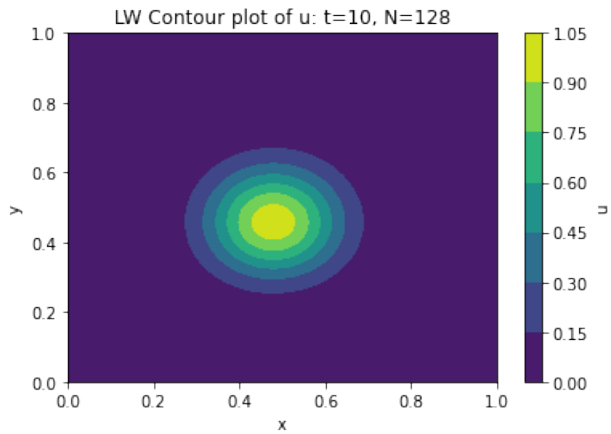


(c)

Figure 7: Lax-Wendroff method for $N = 256$



(a)



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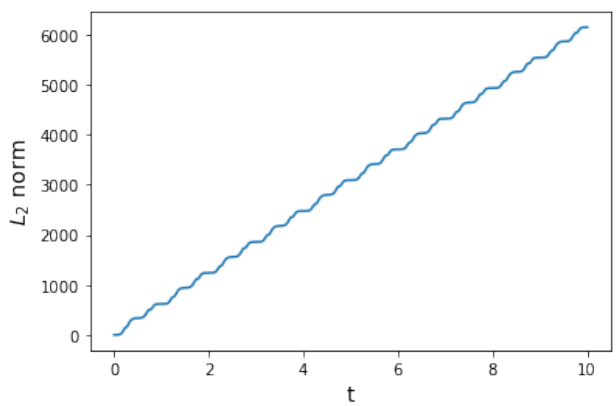
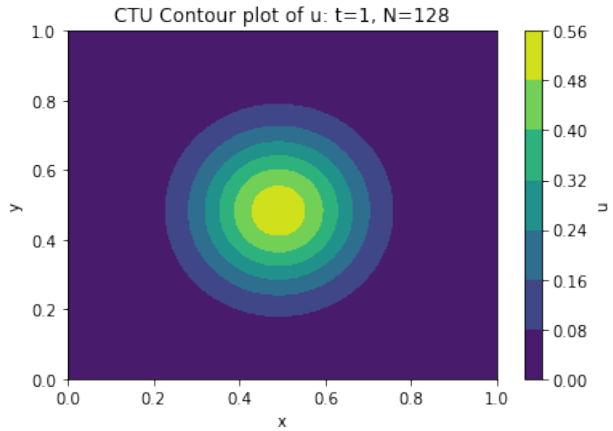
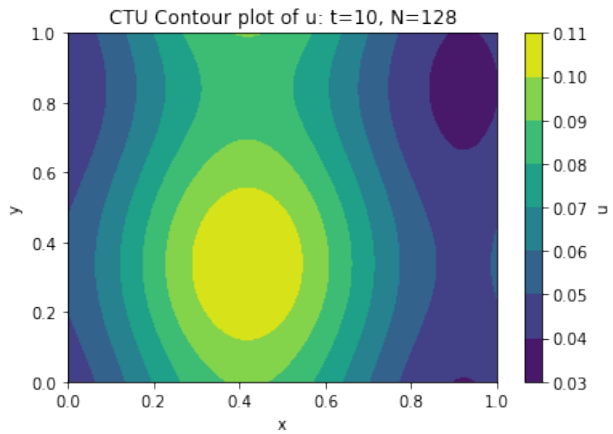


(c)

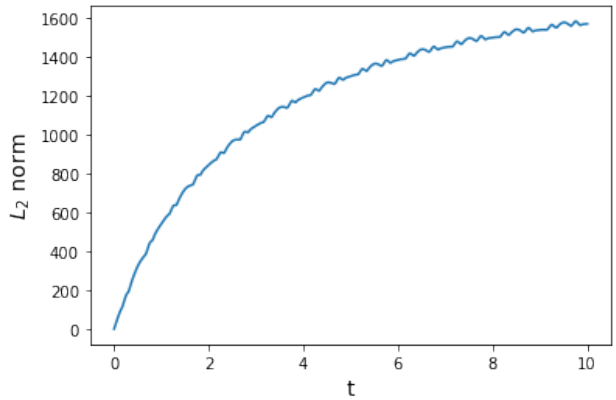
Figure 8: Lax-Wendroff method for $N = 512$



(a)

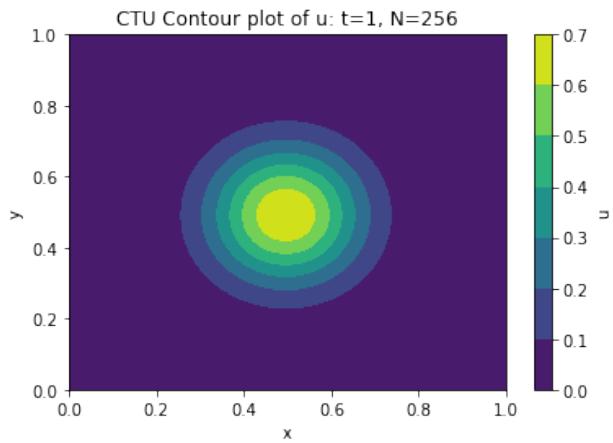


(b)

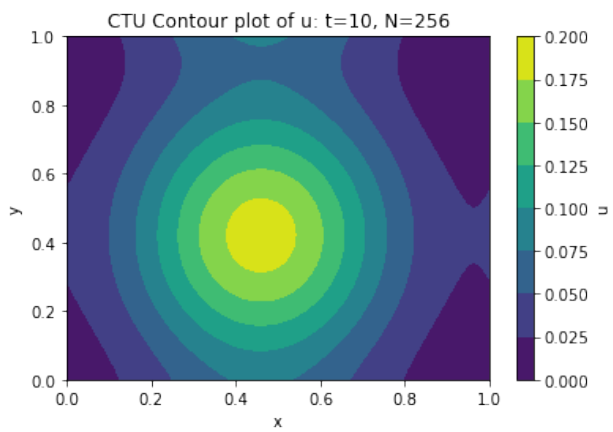


(c)

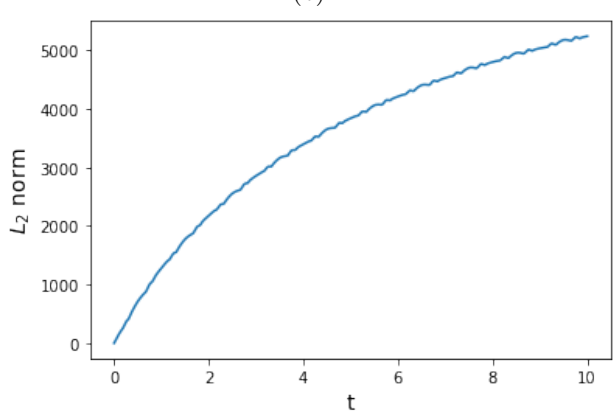
Figure 9: Corner Transport Upstream method for $N = 128$



(a)

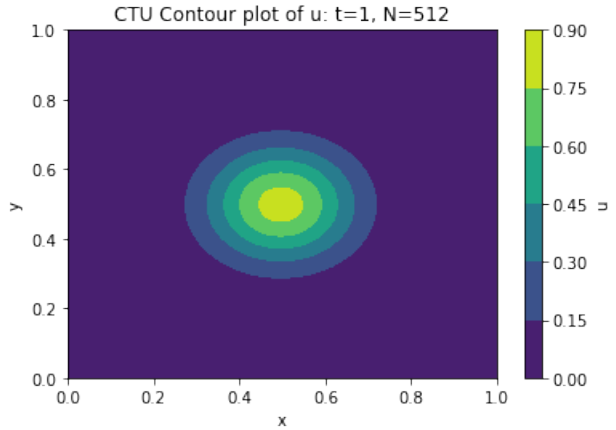


(b)

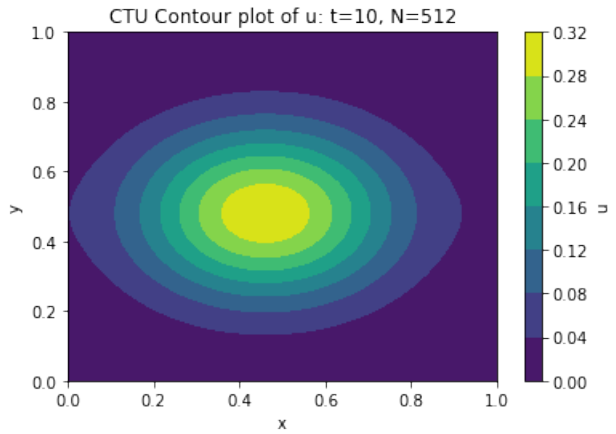


(c)

Figure 10: Corner Transport Upstream method for $N = 256$



(a)



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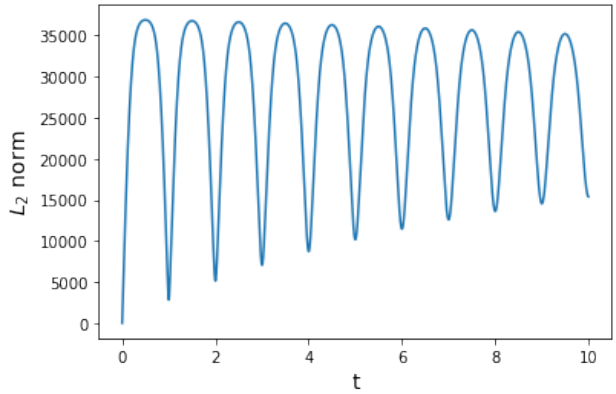

(c)

Figure 11: Corner Transport Upstream method for $N = 512$

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